🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.EE.7

Solve One-Step Addition and Subtraction Equations

Lesson 7-2 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$x + 2 = 7$

balanced with =

$x + 5 = 12$ — both sides equal 12 when $x = 7$

Equation

Write the definition:

$x + 2 = 7$

balanced with =

Addition undoes subtraction: if $x - 3 = 10$, add 3 to get $x = 13$

Inverse operation

Write the definition:

$x + 2 = 7$

balanced with =

$x + 5 = 12$ → subtract 5 from both sides → $x = 7$ (x is isolated)

Isolate

Write the definition:

$x + 2 = 7$

balanced with =

$x = 7$ is the solution to $x + 5 = 12$ because $7 + 5 = 12$ ✓

Solution

Write the definition:

number

$x + 4 = 9$ is solved in one step: subtract 4 from both sides, so $x = 5$.

One-step equation

Write the definition:

number

If you subtract 4 from the left side, you must subtract 4 from the right to keep balance.

Balance

Write the definition:

Guided Notes

Level 2 Standard

🎯 **WHAT WE'RE LEARNING TODAY**

I can solve one-step addition and subtraction equations using inverse operations.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Equation | Inverse operation | Isolate | Solution | One-step equation | Balance

👆 *Tap any word to see what it means and a picture.*

1. A math sentence with an equal sign showing two equal amounts is an __________.

2. An operation that undoes another, like subtraction undoing addition, is an __________.

3. To get the variable alone on one side of the equation is to __________ it.

4. The value of the variable that makes the equation true is the __________.

5. An equation solved in a single step is a __________.

6. Doing the same thing to both sides to keep them equal keeps the equation in __________.

🧮 Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: Solve: $x + 9 = 15$

A. $x = 6$

B. $x = 24$

C. $x = 9$

D. $x = 15$

Step 1 Subtract 9 from both sides: $x = 15 - 9 = 6$.

Step 2 Check: $6 + 9 = 15$ ✓

✅ **Answer:** A. $x = 6$

🧩 **Try — put the steps in order**

Drag the cards (or use the ▲▼ buttons) to put the solution steps in the right order, then press **Check**.

Check: $6 + 9 = 15$ ✓

Subtract 9 from both sides: $x = 15 - 9 = 6$.

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: Solve: $y - 7 = 20$

A. $y = 27$

B. $y = 13$

C. $y = 7$

D. $y = 140$

Step 1 ______________________

Step 2 ______________________

Answer: ______________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Solve: $n + 15 = 42$

A. $n = 27$

B. $n = 57$

C. $n = 15$

D. $n = 42$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Unlock Card 2: Solve n – 4 = 9.

A. n = 13

B. n = 5

C. n = 36

D. n = -5

Show your work:

2. Unlock Card 3: A vault held some clue tokens. After the detective added 16 tokens, there were 50. The equation is t + 16 = 50. How many tokens were there before?

A. 34

B. 66

C. 16

D. 800

Show your work:

Stretch Your Thinking Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A detective solved $x + 25 = 60$ and got $x = 35$. Another detective solved $y - 25 = 60$ and got $y = 35$. Are both correct? Explain using inverse operations and check each answer.

Sentence starter: For $x + 25 = 60$, the inverse operation is ____, so $x =$ ____. Check: ____. For $y - 25 = 60$, the inverse operation is ____, so $y =$ ____. Check: ____. Therefore, ____.

Show your work:

Reflect — Exit Ticket

Solve: $p + 2.8 = 9.1$

A. $p = 6.3$

B. $p = 11.9$

C. $p = 6.8$

D. $p = 3.25$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Equation
2. **Notes 2:** Inverse operation
3. **Notes 3:** Isolate
4. **Notes 4:** Solution
5. **Notes 5:** One-step equation
6. **Notes 6:** Balance
7. **We Try (notes):** A. y = 27 *— Add 7 to both sides: y = 20 + 7 = 27. Check: 27 − 7 = 20 ✓*
8. **You Try (notes):** A. n = 27 *— Subtract 15 from both sides: n = 42 − 15 = 27. Check: 27 + 15 = 42 ✓*
9. **Try It 1:** A. n = 13 *— Add 4 to both sides: n = 9 + 4 = 13. Check: 13 − 4 = 9 ✓*
10. **Try It 2:** A. 34 *— Subtract 16 from both sides: t = 50 − 16 = 34. Check: 34 + 16 = 50 ✓*
11. **Exit Ticket:** A. p = 6.3 *— Subtract 2.8 from both sides: p = 9.1 − 2.8 = 6.3. Check: 6.3 + 2.8 = 9.1 ✓*